

Forecasting the Post-Pandemic Inflation Surge: A Comparative Evaluation of Macroeconomic Models

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1. Introduction and Research Question

The sharp rise in U.S. inflation after COVID-19 created a major challenge for macroeconomic forecasting. Before 2020, inflation was relatively stable, but it rose sharply in 2021 and 2022. This project examines whether conventional forecasting models were able to predict inflation accurately during the post-pandemic period.

The main research question is: What is the most accurate conventional macroeconomic forecasting model for predicting U.S. inflation, and how has model performance changed since 2020?

This question matters because inflation forecasts affect monetary policy, borrowing costs, business planning, and household expectations. If models trained on pre-2020 data perform poorly after 2020, it suggests that the pandemic changed inflation behavior in ways that traditional models did not fully capture.

This project compares four models: a naive model, an autoregressive model, a multivariate regression model, and a vector autoregression model. The analysis uses monthly FRED data, and inflation is calculated as the year-over-year percentage change in CPI.

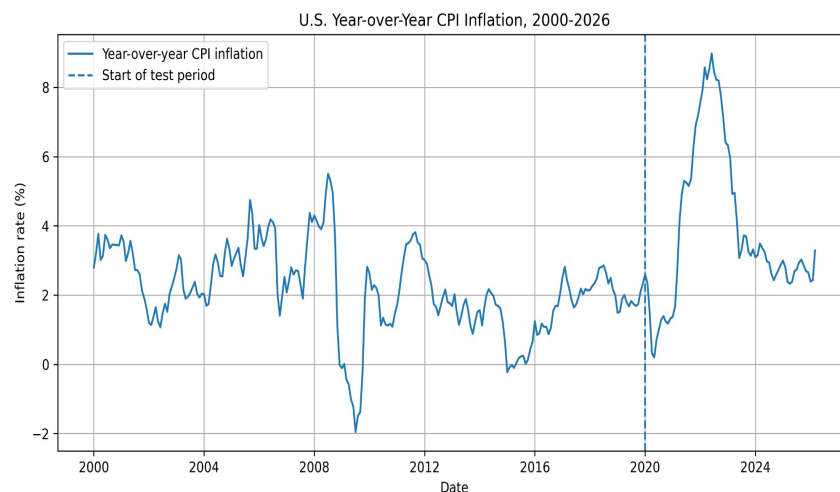


Figure 1. U.S. Year-over-Year CPI Inflation, 2000–2026

Figure 1 shows that inflation was relatively stable before 2020 but increased sharply after the pandemic. This motivates the question of whether models based on pre-pandemic relationships remained accurate after 2020.

2. Brief Literature Review

Inflation forecasting is a central topic in macroeconomics because inflation is persistent, but it is also affected by broader economic conditions. A simple starting point is a univariate or autoregressive model, where future inflation is predicted using current and past inflation. These models are useful because inflation often moves gradually rather than changing randomly from month to month.

Another common approach is based on the Phillips curve, which connects inflation to labor market conditions such as unemployment. If unemployment is low, inflationary pressure may rise; if unemployment is high, inflationary pressure may fall. However, previous research shows that the usefulness of activity-based inflation forecasts can change over time. Stock and Watson find that Phillips curve forecasts sometimes perform better and sometimes worse than univariate benchmarks, meaning that adding variables like unemployment does not always improve forecast accuracy.

This literature motivates the approach in this project. Instead of assuming that a more complex model is automatically better, this project compares simple inflation-based models with models that also include unemployment and interest rates.

3. Data and Methodology

The data come from the Federal Reserve Economic Data database maintained by the Federal Reserve Bank of St. Louis. The project uses monthly observations from January 2000 through March 2026. Three macroeconomic variables are used: the Consumer Price Index for All Urban Consumers, the unemployment rate, and the federal funds effective rate.

Inflation is calculated as the year-over-year percentage change in CPI:

- $$\text{Inflation}_t = ((\text{CPI}_t - \text{CPI}_{t-12}) / \text{CPI}_{t-12}) \times 100$$

CPI is used because percent changes in the CPI measure inflation, and FRED describes the percent change from one year ago as the most common inflation metric. The unemployment rate is included to capture labor market conditions, while the federal funds rate is included to capture monetary policy conditions.

The forecasting target is next month's year-over-year inflation. The main models are trained using data from 2000 to 2019 and tested on the period from 2020 to 2026. This setup evaluates how well models estimated before the pandemic performed during the post-pandemic inflation surge.

Four forecasting models are compared:

- 1. Naive model: next month’s inflation equals current inflation.
- 2. Autoregressive model: next month’s inflation is predicted using current and lagged inflation.
- 3. Multivariate regression model: inflation is predicted using inflation history, unemployment, and the federal funds rate.
- 4. Vector autoregression model: inflation, unemployment, and the federal funds rate are modeled jointly.

Forecast accuracy is evaluated using mean absolute error (MAE) and root mean squared error (RMSE). Lower MAE and RMSE indicate better performance. RMSE is emphasized because it penalizes large forecast errors more heavily.

4. Results

Table 1 shows the main forecast accuracy results for the 2020–2026 testing period. The autoregressive model performs best, with the lowest MAE and RMSE. The naive model is close behind, while the multivariate regression and VAR models perform worse.

	Model	MAE	RMSE
0	Autoregressive Model	0.330908	0.433864
1	Naive Model	0.343224	0.463410
2	Multivariate Regression Model	0.374142	0.499515
3	VAR Model	0.451186	0.642726

Table 1. Forecast Accuracy During the Post-2020 Test Period

The results suggest that recent inflation history was more useful than unemployment and interest rates for forecasting post-2020 inflation. The weaker performance of the multivariate model suggests that pre-2020 relationships did not transfer well to the post-pandemic period.

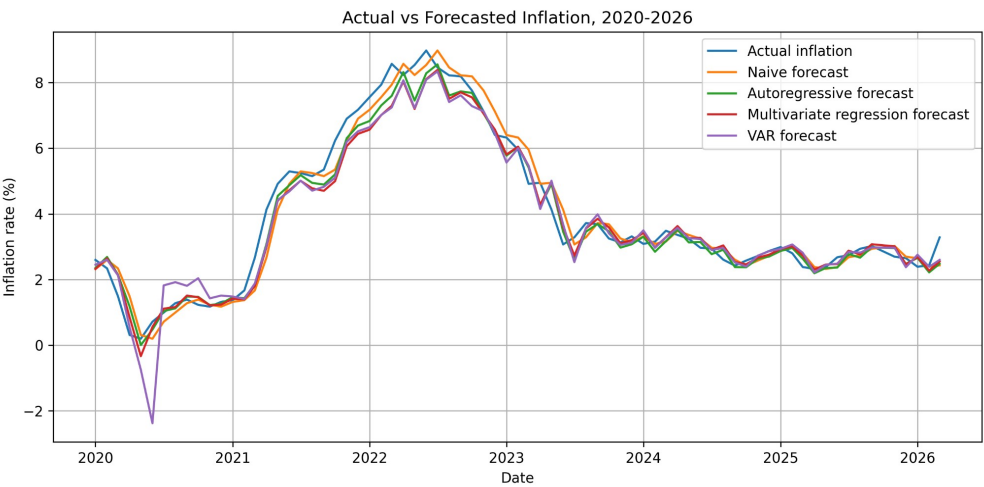


Figure 2. Actual vs Forecasted Inflation, 2020–2026

Figure 2 shows that the models generally follow the broad direction of inflation, but they struggle during the sharp rise and decline in inflation between 2021 and 2023. This supports the idea that the post-pandemic inflation surge was difficult for conventional

models to forecast.

5. Did Forecasting Performance Change After 2020?

To test whether inflation became harder to forecast after 2020, model performance is compared across two periods: 2015–2019 and 2020–2026. The 2015–2019 period acts as a pre-pandemic validation period, while 2020–2026 captures the post-pandemic inflation surge.

Table 2 shows that forecast errors increased in the post-2020 period for every model. Before the pandemic, all four models had RMSE values near 0.26. In the post-2020 period, RMSE increased substantially, especially for the VAR model.

	Base Model	Period	MAE	RMSE
0	Naive Model	2015-2019	0.203250	0.260906
1	Autoregressive Model	2015-2019	0.201204	0.262687
2	Multivariate Regression Model	2015-2019	0.200707	0.267486
3	VAR Model	2015-2019	0.210148	0.264778
4	Naive Model	2020-2026	0.343224	0.463410
5	Autoregressive Model	2020-2026	0.330908	0.433864
6	Multivariate Regression Model	2020-2026	0.374142	0.499515
7	VAR Model	2020-2026	0.451186	0.642726

Table 2. Forecast Accuracy Before and After 2020

The increase in RMSE shows that forecasting inflation became more difficult after 2020. The autoregressive model remained the strongest post-2020 model, but its RMSE was still much higher than in the pre-pandemic validation period.

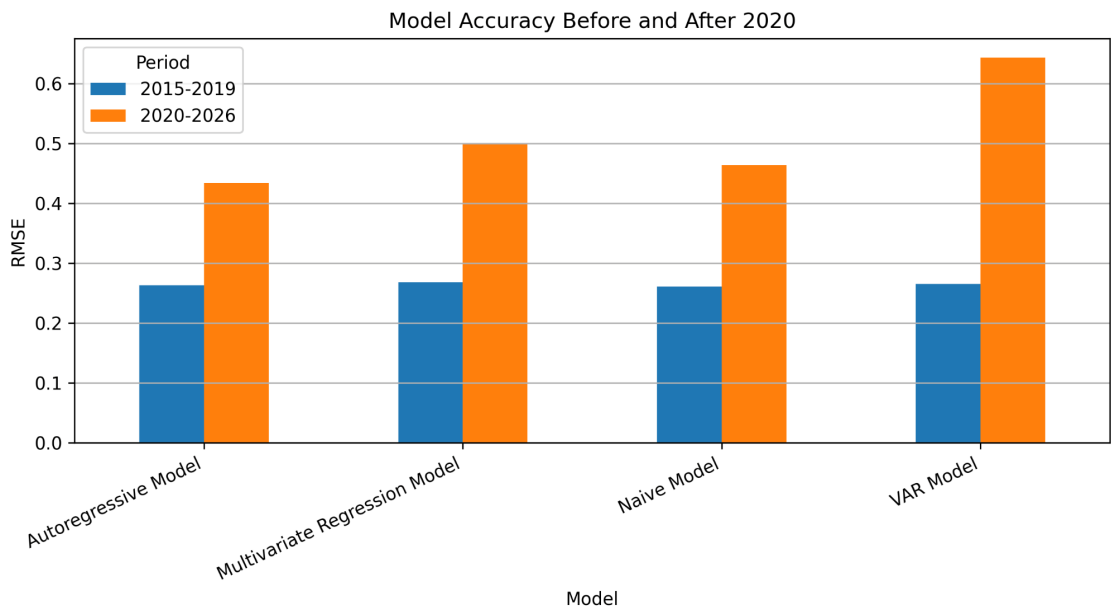


Figure 3. Model Accuracy Before and After 2020

Figure 3 visually confirms the result in Table 2. All models have higher RMSE after 2020, indicating that the post-pandemic inflation surge reduced forecast accuracy.

6. Conclusion and Extensions

This project compared four conventional macroeconomic models for forecasting U.S. inflation after 2020. The main finding is that the autoregressive model performed best during the 2020–2026 testing period, with the lowest RMSE. This suggests that recent inflation history was more useful for predicting post-pandemic inflation than adding unemployment and federal funds rate variables.

The results also show that inflation became harder to forecast after 2020. In the 2015–2019 validation period, all models had RMSE values near 0.26. In the 2020–2026 period, forecast errors increased for every model. This supports the conclusion that the post-pandemic inflation surge weakened the accuracy of conventional forecasting models.

One possible explanation is that the post-pandemic period included unusual shocks that were not captured well by the models, such as supply chain disruptions, fiscal stimulus, energy price changes, and rapid monetary policy adjustments. These factors may have changed the relationship between inflation, unemployment, and interest rates.

Future extensions could include additional predictors such as oil prices, money supply growth, supply chain pressure indexes, or inflation expectations. Another extension would be to use rolling-window forecasts so that the models update more gradually over time. The project could also compare headline CPI inflation with core CPI inflation to see whether food and energy prices played a major role in forecast errors.

7. References

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